The Noisy Work of Uncertainty Visualisation

Harriet Mason
Department of Econometrics and Business Statistics, Monash University
and
Dianne Cook
Department of Econometrics and Business Statistics, Monash University
and
Sarah Goodwin
Department of Human Centred Computing, Monash University
and
Emi Tanaka
Biological Data Science Institute, The Australian National University
and
Susan VanderPlas
Statistics Department, University of Nebraska–Lincoln

March 18, 2026

Abstract

Better representation of the uncertainty in a data visualisation is a focus of recent research activity. A problem with the current literature is that there is a lack of clarity about the definition of uncertainty and what it means to represent it in a plot. This confusion results in a significant amount of conflicting results in the literature, especially in experiments that assess the effectiveness of different uncertainty representations. In this review, we summarise the current literature, provide workable definitions, and illustrate these definitions with examples. In doing so, we ask what it really takes to achieve transparency in statistical graphics. It is hoped that it will be useful for guiding new graphics methodology and experimental research.

1 Introduction

What do we mean when we talk about "uncertainty visualisation"? The phrase can feel like a bit of a contradiction to anyone who is familiar with the term. Among statisticians, "uncertainty" is often discussed as though it is an omnipresent spectre that touches every stage of our analysis without ever being fully seen. Authors will often mention that the phrase is incredibly vague (Spiegelhalter 2017, Griethe & Schumann 2006), or talk around the topic by describing a list of things uncertainty *could* be (Kinkeldey et al. 2014, Hullman 2016), but rarely do we see authors attempt to discuss what uncertainty actually *is*. Almost as if in direct contrast with this, visual statistics (information visualisations, data plots) are one of the most powerful tools in the statistician's toolbox, allowing for quick and memorable communication that identifies quirks in our data that we didn't even know to look for. We see this in datasets such as Anscombe's quartet (Anscombe 1973) or the Datasaurus Dozen (Matejka & Fitzmaurice 2017, Locke & D'Agostino McGowan 2018), where visual statistics are able to highlight elements of the data that are invisible to the typical summary statistics. We also see this in recall experiments, where something as simple as sketching a distribution before recalling statistics or making predictions can greatly increase the accuracy of those measures [Hullman et al. (2018); Goldstein2014]. Taken together, uncertainty visualisation implies a need to pull back the curtain and expose put light on the unknowns of our analysis.

As nice as this sentiment is, it turns out to be easier said than done. Reviews on uncertainty visualisation rarely offer tried and tested rules for effective uncertainty visualisation, but rather, they comment on the difficulties faced when trying to summarise the field. Kinkeldey et al. (2014) found most experimental methods to be ad hoc, with no commonly agreed upon methodology, formalisations, or a greater goal of describing general principles. Hullman (2016) noticed there is a serious noise issue in the field, with noise from participants misun-

derstanding visualisations, misinterpreting questions, and incorrectly applying heuristics, all overwhelming any information we can glean from studies. MacEachren et al. (2005) identified so much contradicting evidence that they were able to spend an entire page discussing the conflicting evidence for the question: "Should I map uncertainty to colour hue?" Spiegelhalter (2017) concluded that different plots are good for different things, and ultimately disagreed with the goal of identifying a universal best plot for all people and circumstances. Padilla, Kay & Hullman (2022) summarised several cognitive effects that repeatedly arise in uncertainty visualisation experiments; these effects were each discussed in isolation as a list of considerations an author might make, rather than an overarching theory of rules for effective uncertainty visualisation.

While these reviews are thorough in scope, none discuss how the existing literature contributes to the broader goal of uncertainty visualisation. The problem faced by the literature is easily summarised with a famous quote by Henri Poincaré.

> "Science is built up of facts, as a house is built of stones; but an accumulation of facts is no more a science than a heap of stones is a house." - Henri Poincaré (1905)

That is to say, despite the wealth of reviews, the field of uncertainty visualisation remains a heap of stones. There is a mountain of work that identifies common heuristics found in uncertainty visualisations, evaluates competing plot designs, or starts a theoretical discussion on a niche aspect of the field. This is important work that needs to be done, but each of these papers offers up its own bespoke motivation and methodology, with little reference to the uncertainty visualisation papers outside their periphery. This becomes even more difficult to manage when these studies conflict. The field is in desperate need of a unifying theory that can tie this swath of research together. This review attempts to address this

issue by offering a novel perspective on the uncertainty visualisation problem. That is, we are going to use the wealth of established stone to construct the foundations on which we can build a house.

2 The purpose of uncertainty visualisation

Mentions of "uncertainty visualisation" start springing up around 1990, across several different fields (Ibrekk & Morgan 1987, MacEachren 1992), each with its own motivation for the work. In computer science the area appears to be motivated by issues in the public's perception of random variables, with the hope that visualisations would give laypeople the ability to extract important information from graphical representations (Ibrekk & Morgan 1987). With similar concerns about the public's understanding of randomness, the field of psychology, statistics, and economics used "uncertainty visualisations" as a communication tool to mitigate the psychological bias associated with the communication of risk, a topic of concern since the early 1980s (Spiegelhalter 2017). In the cartography literature, it was motivated by inherent uncertainty of geoscience data, the practical use of visualisation as an exploratory tool, and the constrained visual channels from map representations (MacEachren 1992). These disparate motivations have merged together, and today, uncertainty visualisation is usually motivated by the vague goal of "decision-making". This term has been used to mean the mitigation of psychological bias to ensure economically rational decisions (Padilla, Hosseinpour, Fygenson, Howell, Chunara & Bertini 2022, Padilla et al. 2021, Kale et al. 2021), the facilitation of trust or confidence (Zhao et al. 2023, Yang et al. 2023), the ability to extract values related a distribution (Sarma et al. 2023), the prevention of false discovery in plots (Sarma et al. 2024, Koonchanok et al. 2023); or the extraction of some other metric that is vaguely related to "uncertainty" (Chakraborty et al.

2024, Ndlovu et al. 2023). This gradual scope creep of the field, motivated by the hazy definition of "decision-making", is the most likely culprit for the irreconcilable literature that makes up the field today.

Given that there is so much subliminal disagreement in uncertainty visualisation, how did these disparate motivations come to be seen as interchangeable? All discussions on uncertainty visualisation seem to have a common thread that connects them; the belief that the ultimate goal of uncertainty visualisation is not trust, rationality, or value extraction, but *transparency*. We see it said directly in reviews of the field (Padilla, Kay & Hullman 2022). We see it when authors claim that failing to include uncertainty is akin to fraud or lying (Hullman 2020, Manski 2020). We see it when authors assert that uncertainty communicates the legitimacy (or illegitimacy) of the conclusion drawn from visual inference (Correll & Gleicher 2014, Kale et al. 2018, Griethe & Schumann 2006). We see it when authors say uncertainty visualisations should communicate a degree of confidence (Boukhelifa et al. 2012) or validity (Hullman 2020, Griethe & Schumann 2006) in our conclusions. We see it when authors suggest uncertainty visualisation should "guide, qualify, or soften our judgements of uncertain data" (Wilkinson 2005). These authors are not wrong about the need for transparency in science communication, either. A 6-month survey of anti-mask groups on Facebook during the COVID-19 pandemic showed that anti-maskers made persuasive arguments by exploiting inherent uncertainty that was ignored by pro-maskers (Lee et al. 2021).

This consensus leads us to understand that uncertainty visualisation is motivated by the need for a sort of "visual hypothesis test", a sentiment expressed by some authors directly (Correll et al. 2018, MacEachren 1992). A successful uncertainty visualisation would act as a "statistical hedge" for any inference we make using the graphic. Since the purpose of a

visualisation is to give a quick gist of the information (Spiegelhalter 2017), this hedging should be communicated visually without the need for complicated mental calculations. Therefore, an effective uncertainty visualisation should not just "show" uncertainty; untrustworthy conclusions should *not be visible*. If we refer to the conclusion we draw from a graphic as its signal and the uncertainty that should make this signal harder to read as the "noise" increases, we can summarise the above information into three key requirements. A good uncertainty visualisation should:

1) Reinforce justified signals to encourage confidence in results.
2) Hide spurious signals that are overwhelmed by noise.
3) Perform tasks 1) and 2) in a way that is proportional to the level of confidence in those conclusions.

Usually, visualisations that are unconcerned with uncertainty have no issue showing justified signals, but struggle with the display of unjustified signals. Therefore, we *suggest calling this approach to uncertainty visualisation "signal-suppression"* since it primarily differentiates itself from the normal "noiseless" visualisation approach through criterion (2). This is the main criterion we will use to assess the current literature on uncertainty visualisation.

3 Current Approaches

3.1 Ignoring uncertainty

The most common way to visualise uncertainty is to simply not do it at all. A study conducted by Hullman (2020) found that only a quarter of authors surveyed included uncertainty in 50% or more of their visualisations. Part of the abstinence comes from the fact that authors are not sure how to calculate the uncertainty (Hullman 2020). This is not

entirely unreasonable, given that even visualisation authors themselves seem to be in conflict about what exactly uncertainty is. Therefore, it makes sense to start with visualisations that ignore uncertainty with the hope that, by looking at where uncertainty *isn't*, we can better understand where it *is*. (XXX don't understand the secon part of this statement)

3.1.1 What is uncertainty?

It is surprisingly hard to describe what uncertainty actually is. Most authors simply avoid the problem and describe the characteristics of uncertainty, of which there are plenty. Often, uncertainty is split using an endless stream of ever changing boundaries, such as whether the uncertainty is: due to true randomness or a lack of knowledge (Begg et al. 2014, Spiegelhalter 2017, Gustafson & Rice 2019, Padilla, Kay & Hullman 2022, Hullman 2016, Walker et al. 2003) quantifiable or unquantifiable (Spiegelhalter 2017, Walker et al. 2003, Padilla, Kay & Hullman 2022); scientific or human (Benjamin & Budescu 2018, Gustafson & Rice 2019); systematic or random (Sanyal et al. 2009); statistical or bounded (Gschwandtner et al. 2016, Olston & Mackinlay 2002); accuracy or precision (Griethe & Schumann 2006, Benjamin & Budescu 2018, Hullman 2016); etc. There are enough qualitative descriptors of uncertainty to fill a paper, but none of this is particularly helpful in understanding how to integrate it into a visualisation.

Rather than trying to define uncertainty by looking at the myriad ways in which it *does* appear in an analysis, we may find it easier to look at where it *does not*. Descriptive statistics describe our sample as it is and summarise large data down into an easy-to-swallow format. Descriptive statistics are not seen as the primary goal of modern statistics; however, this was not always the case. In 19th-century England, *positivism* was the popular philosophical approach to science (positivists included famous statisticians such as Francis Galton and

Karl Pearson). Practitioners of the approach believed statistics ended with descriptive statistics as science must be based on actual experience and observations (Otsuka 2023). In order to make statements about population statistics, future values, or new observations, we need to perform inference, which requires the assumption of the "uniformity of nature", that is, we need to assume that unobserved phenomena should be similar to observed phenomena (Otsuka 2023). Positivists believed referencing the unobservable was bad science. In other words, these scientists embraced descriptive statistics due to the inherent certainty that came with them. Since uncertainty is nonexistent in descriptive statistics, it is clear that uncertainty is a by-product of inference. Uncertainty is the noise that is both inseparable from our inference and meaningless without it.

If we consider uncertainty as a byproduct of statistical inference, it would make sense to simply quantify it as a distribution. Suggesting distributions as a representation of uncertainty is not new. Kay (2023) originally suggested thinking about uncertainty visualisations as visualisations with distribution inputs, to replace the commonly used mean and standard deviation, removing the assumption of a Gaussian distribution. Taking this point a step further, we suggest that a distribution is sufficient to characterise *any* uncertainty, no matter how intangible it is. This is in contrast to the assertions made by some authors, who believe abstract notions of uncertainty, such as credibility (Thomson et al. 2005), forecaster confidence (Padilla et al. 2021), or uncertainty about uncertainty (Hadjimichael et al. 2024), are too complex to be quantified. This is simply untrue, as abstract concepts such as human belief or credibility are regularly quantified by Bayesians and hierarchical models are often used for uncertainty about uncertainty.

3.1.2 Example: ignoring uncertainty

If visualising uncertainty is fundamentally a scenario where we are visualising a set of random variables, what does "ignoring" uncertainty look like? Figure 1 shows six different plots, a high and low uncertainty case for three different scenarios, all of which visualise the expected value of the input distribution. Plots R1 and R2 visualise the line from a simple linear regression on the `mtcars` data, with differing sample size underlying the uncertainty. Plots S1 and S2 show a choropleth map, where counties in Iowa are coloured according to a simulated temperature measurement, with measurement error being the uncertainty. Plots G1 and G2 show a summary of measurements on the same variable in five populations. In G1 distributions have low variance, and in G2 they have high variance. In the linear regression, we see a downward trend, and because this is overlaid on the observed data we can get a sense of the uncertainty. It is not ignored in this plot, because the data is plotted. In the map you should see a sine wave spatial trend. In the univariate distributions, an incremental increase in treatment. In S1, S2, G1, G2 what we cannot see is that the variance on the estimates in the bottom row is significantly higher than that in the top row, so much so that any visible trend should actually disappear. If we were to ask a reader: Can you see a difference between the plots in the top row versus the bottom row? Is the strength of the trend communicated through the visualisation? The answer to both of these questions will be no, as the high and low variance cases are identical.

We will use these plots as a touchstone for our discussion on uncertainty visualisation, and return to them when we discuss the approaches suggested in the literature. This example will be used to assess the approach on how well suggested methods hide spurious relationships and reveal the hidden variance structures in the data.

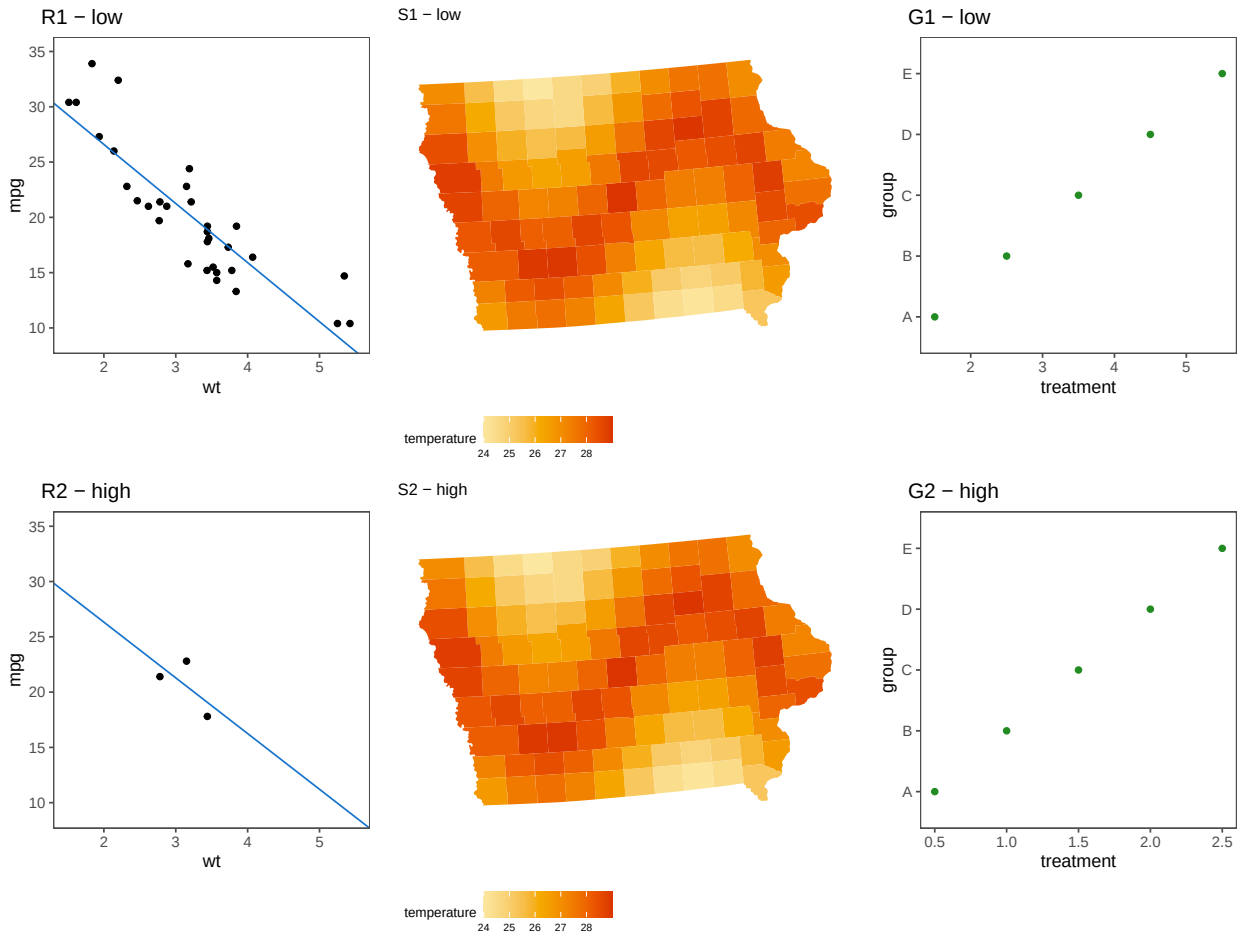

Figure 1: Three example types of data and associated plots that will be used to illustrate various choices of uncertainty representation throughout the paper: scatterplot and regression line (R1, R2), spatial choropleth map (S1, S2), grouped dotplot (G1, G2). Two levels of uncertainty (low, high) is used with each example. Only with R1 and R2, where the regression overlays the scatterplot can we see a difference between low and high.

3.2 Uncertainty as a statistic

The most common approach to uncertainty visualisation is to swap out our original statistic for one that shows uncertainty. We see it as an exceedance probability map in geoscience (Kuhnert et al. 2018, Lucchesi et al. 2021), as an icon array in risk communication (Spiegelhalter 2017), or as a summary plot in computer science (Potter et al. 2010). Some authors take this approach because they explicitly believe uncertainty is a variable of importance in and of itself (Blenkinsop et al. 2000), others straddle the line, asserting uncertainty is acting as signal and noise and should fulfil both roles (Peña-Araya et al. 2025). This seems a little too easy. Can we just swap out the statistic for an "uncertainty statistic" and get an "uncertainty visualisation"? Is it that easy?

3.2.1 Example: visualising variance

Figure 2 depicts the six plots from Figure 1, except that the original central value estimate has been replaced with an uncertainty statistic. Plots (a) and (b) are the most absurd examples of this visualisation approach, as replacing the estimated slope and intercept of a line with the standard error of said estimate produces a plot that is inherently meaningless. This will not always be the case, as we can see in the exceedance probability maps depicted in plots (c) and (d) that visualise $P(Temperature > 27)$ for each county. Plots (e) and (f) simply show a scatter plot that depicts the variance of each distribution instead of the mean. In terms of the actual geometric object used to visualise our data, these graphics plots are identical to their certain counterparts. While it is only the statistic that has changed, this has changed the fundamental meaning of the plot. This change in the meaning results in the trend of the regression, and univariate distributions becoming invisible, but that is because we are visualising a fundamentally different plot, rather than a proper integration

of uncertainty.

3.2.2 What is an uncertainty visualisation?

The question about what is, or is not, an uncertainty visualisation seems to be one of the most pervasive divides in the literature. For example, Wilkinson (2005) mentions that popular graphics, such as pie charts and bar charts, omit uncertainty, and Wickham & Hofmann (2011) suggests their product plot framework, which includes histograms and bar charts, should be extended to include uncertainty. However, pie charts, bar charts and histograms have all been used as a significant number of experiments as examples of an "uncertainty visualisation" (Ibrekk & Morgan 1987, Olston & Mackinlay 2002, Zhao et al. 2023, Hofmann et al. 2012). What is going on here?

This conflict relates to the subconscious disagreement in the purpose of uncertainty visualisation. If you believe uncertainty visualisation is about communicating risks or random variables, uncertainty visualisations are just visualisations of "uncertainty statistics" or distributions. On the other hand, if you believe uncertainty visualisation is about suppressing false signals in a graphic, then you see an uncertainty visualisation as a transformation of an existing graphic that somehow adds the uncertainty in. The former has no limitation on the visual appearance of an "uncertainty visualisation", allowing pie charts, bar charts or histograms, so long as the graphic is visualising "uncertainty", while the latter believes uncertainty visualisations only exist in relation to some "normal" visualisation. When we refer to the graphics depicted in Figure 2 as "uncertainty visualisations", we are classifying visualisations by the data they display, not their visual features. This is not the standard approach in statistical graphics. A scatter plot that compares means and a scatter plot that compares variances are both scatter plots.

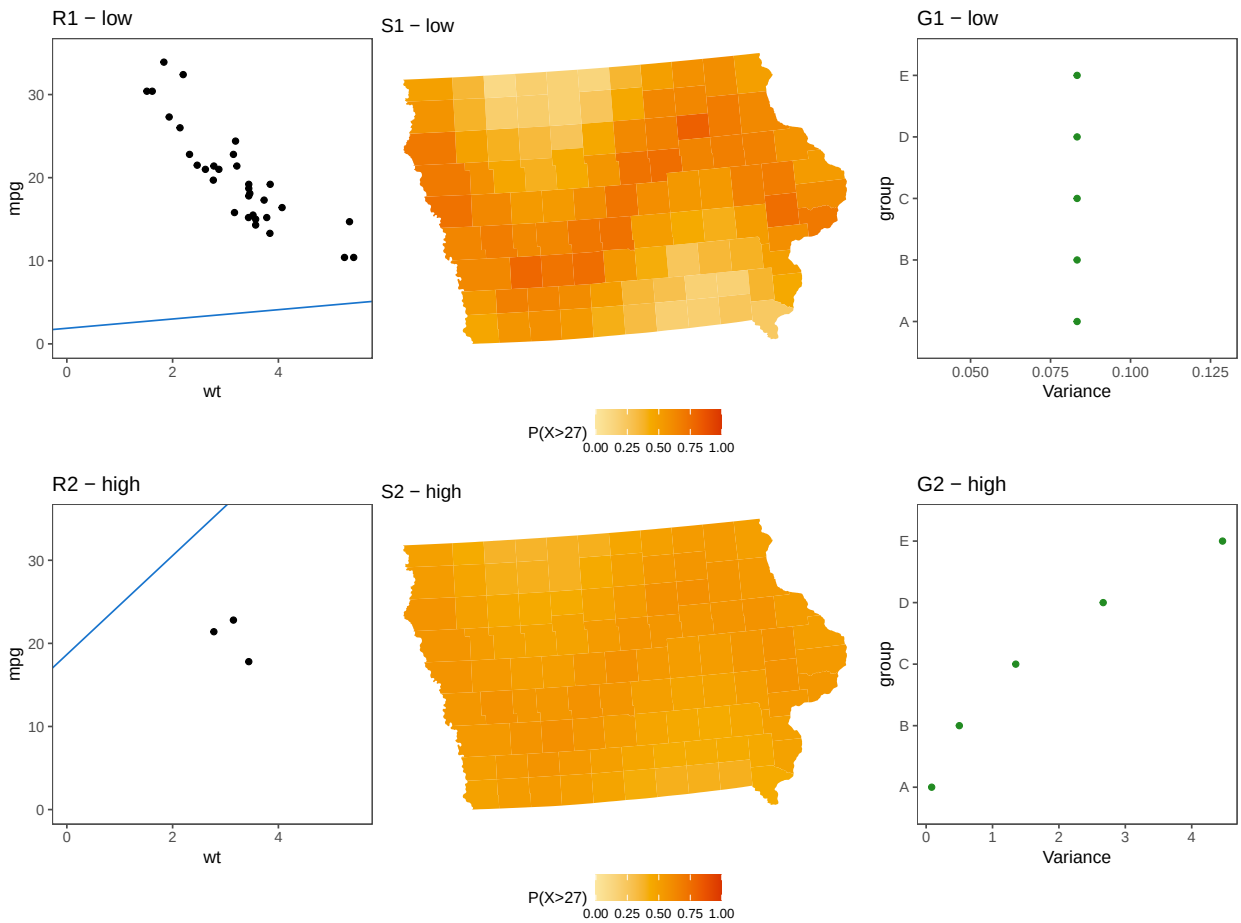

Figure 2: Six different plots, each depicting a high and low uncertainty case for three plots (regression, choropleth, and group comparison) commonly seen in discussions around uncertainty visualisation. Plots (a) and (b) have the standard error of the slope and intercept mapped to the slope and intercept of the line, plots (c) and (d) visualise the estimated value of P(X>27), and plots (e) and (f) visualise the mean of each group. We can see that there are some differences between the top and bottom row, but our visualisations are functionally meaningless in their new context. The standard error of a slope and intercept has no meaning when mapped to a line; the spatial pattern in the high and low spatial map is still visible. The group comparison is interpretable, but we are looking at a fundamentally different visualisation.

Unlike plots, which are not defined by our statistic, uncertainty is *only* defined in relation to our visual statistic, a topic that frequently appears in the literature as a dependence on the "goals" of our analysis. Meng (2014) commented that what is kept as data and what is tossed away is determined by the motivation of an analysis, as what was previously noise can be shown to become signal depending on the question we seek to answer.

Otsuka (2023) suggested that the process of observing data to perform statistics is largely dependent on our goals, because the process of boiling real-world entities down into probabilities depends on the relationship we seek to identify with our data. Wallsten et al. (1997) argue that the best method for evaluating or combining subjective probabilities depends on the uncertainty the decision maker wants to represent and why it matters. Fischhoff & Davis (2014) suggested we should have methods for communicating uncertainty depending on what the user is supposed to do with it.

Spiegelhalter (2017) noted we "cannot assess the quality of risk communication unless the objectives are clear". Peña-Araya et al. (2025) asserted that whether or not it is a source of doubt depends on the context. The sentiment behind this repeated point is clear: the role of uncertainty or signal is not dependent on the "type" of statistic, on the source of the information, or the methods we use; it is determined by the statistic we wish to draw inference on. Therefore, the fundamental problem with the "uncertainty statistic" approach is that the uncertainty in the plot isn't acting as noise; it is acting as signal.

As the uncertainty in this graphic is acting as a signal, there isn't an interesting perceptual challenge associated with the visualisation, and they should just be visualised using existing principles of graphic design. In changing the inferential statistic, we also haven't dealt with the original problem of integrating noise, as these "uncertainty statistics" *also have associated uncertainty in the estimates* that we are ignoring in this visualisation. There is

nothing wrong with explicitly visualising variance, error, bias, or any other statistic used to depict uncertainty as a signal. Just like any other statistic, these metrics provide important and useful information for analysis and decisions. The problem with this approach is that it means everything is an uncertainty visualisation, and if everything is an uncertainty visualisation, nothing is.

3.3 Uncertainty as a variable

Another common sentiment in the literature is the characterisation of uncertainty as just another variable that needs to be integrated into the visualisation, which means uncertainty visualisation is, at its core, a high-dimensional visualisation problem (Griethe & Schumann 2006, Peña-Araya et al. 2025). This sentiment is quite common, with many discussions across computer science (Kinkeldey et al. 2014), cartography (MacEachren et al. 2005), and statistical graphics (Wilkinson 2005) treating this "additional aesthetic" approach as the only method.

This approach is often discussed in relation to how "integrated" the uncertainty is with the estimate. Kinkeldey et al. (2014) identified a split between intrinsic plots, where we map uncertainty to the colour or size of the geometric object of our estimate, and extrinsic plots, where uncertainty is mapped to a separate geometric object, such as glyphs or error bars. A similar distinction was made by Padilla, Kay & Hullman (2022), who classified uncertainty visualisations as graphical annotations (extrinsic), and probability mapped to a visual encoding channel (intrinsic), or a hybrid of the two. It is unclear if these levels of "integration" in a plot design affect its ability to perform signal suppression.

3.3.1 Example: mapping two independent variables

Figure 3 depicts the six visualisations Figure 1, except we have also mapped uncertainty to a spare aesthetic in the plot. The integrable approach is shown in plots (a) and (b), where the standard error of the slope is represented by the width of the line, and the standard error of the intercept is mapped to transparency. A bivariate map is shown in plots (c) and (d), and it offers a slight divergence from the intrinsic/extrinsic approach as it maps the estimate to colour hue and the variance to colour saturation, keeping the signal and noise contained to one visual aesthetic. The extrinsic approach, shown in plots (e) and (f), where we have represented the estimate with a point, and represented the variance as a 95% confidence interval, with a line range. While the high and low variance cases do not look identical as they did in Figure 1, the trend is still visible in the bottom row, where the variance is high, thus failing at signal suppression. Because these graphics visualise the distribution's estimate and uncertainty as two separate pieces of information, the message of the plot is two separate pieces of information that say "here is the trend *and* here is the uncertainty". It is worthwhile to interrogate why this occurs to see if we can move towards a version of this plot where we are able to communicate signal and noise as a single signal.

3.3.2 How to visualise a "single integrated uncertain value"

The reality is, based on our discussion on inferential statistics, uncertainty *isn't* a separate variable; it is a component of the random variable that is indistinguishable from the random variable itself. Similarities between the "high-dimensional" approach and the "uncertainty as a statistic" approach are apparent when we read motivations for using this method over visualising an estimate (i.e. Figure 1) and variance (i.e. Figure 2) side by side on separate graphics. The preference for the methods utilised in Figure 3 is usually motivated

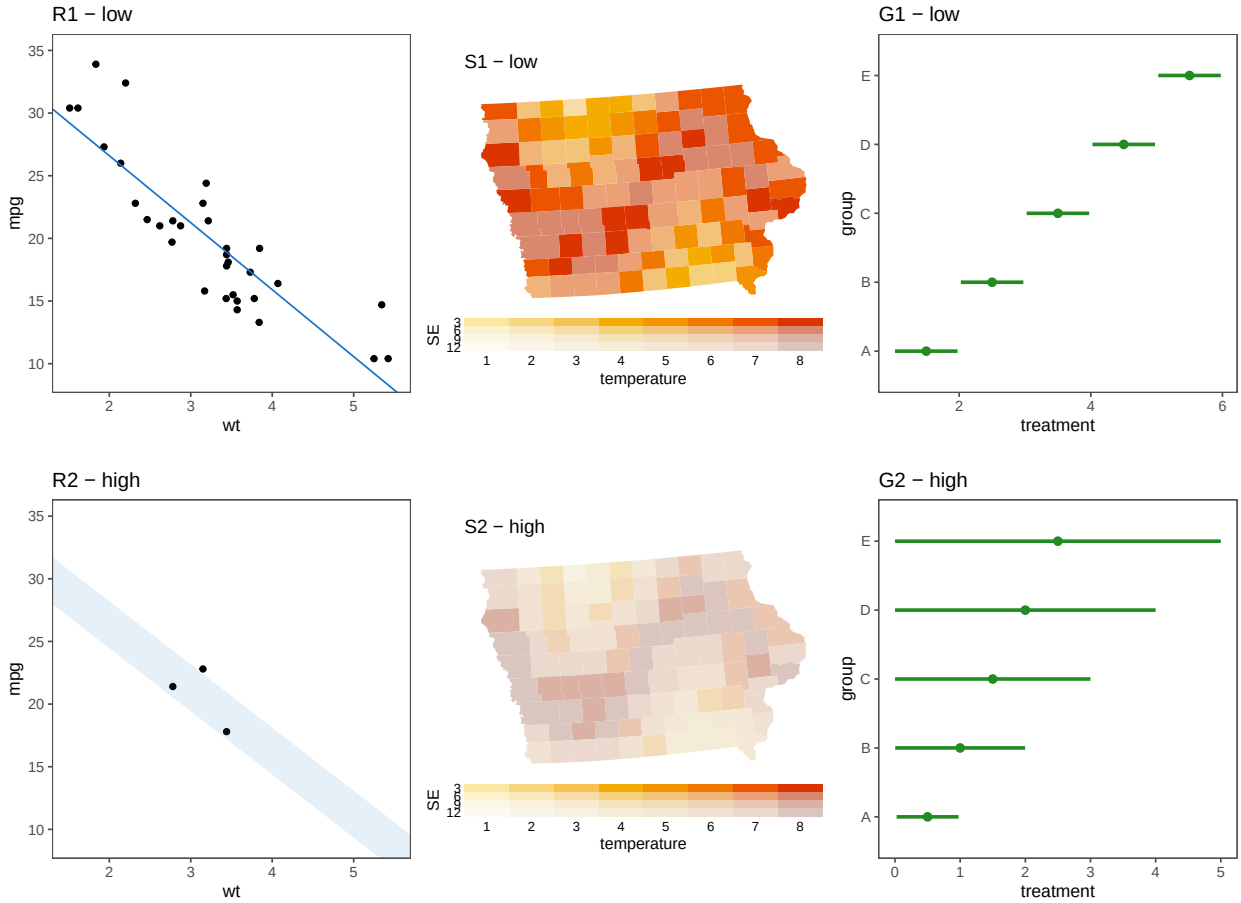

Figure 3: Six different plots, each depicting a high and low uncertainty case for three plots (regression, choropleth, and group comparison). Plots (a) and (b) have the standard error of the slope mapped to the line width, while the standard error of the intercept is mapped to the transparency. Plots (c) and (d) use a bivariate colour palette where the mean of the distribution is mapped to the colour hue, while the variance is mapped to the colour saturation. Plots (e) and (f) visualise the mean of each group as a point, and the uncertainty as an interval. We can see that there are some differences between the top and bottom row, but the 'signals' in the plots are still visible in the bottom row.

by the difficulties in combining information on two separate graphics [Moritz et al. (2017); Correll2018], rather than an understanding that the "uncertainty statistic" approach is not philosophically sound. To achieve signal suppression, we need to visualise noise and signal together as a "single integrated uncertain value" (Kinkeldey et al. 2014) rather than as two separate statistics.

Just because we can still see the signal in Figure 1, that does not mean the reading of the estimate is completely independent of the estimate. When making "non-uncertainty" visualisations, we usually want the visual channels to be separable, that is, we don't want the data represented through one visual channel to interfere with the others (Smart & Szafir 2019). While this approach works for standard data visualisation, in uncertainty visualisation, it allows the estimate and its variance to be read independently, leading to the uncertainty being ignored (Padilla, Kay & Hullman 2022). Therefore, rather than trying to maintain visual separability, the goals of uncertainty visualisation align far better with the pursuit of visual integrability. In an ideal system, our estimate and uncertainty would be separately manipulable, but so well integrated that they are read as a single channel by the human brain. The problem is that even if we implement the most extreme versions of integrable, our methods fall short, as illustrated by the bivariate map in Figure 3. Colour hue and brightness are one of the classic examples of integrable variables (Vanderplas et al. 2020), and decreasing saturation should make the colours harder to distinguish, but the signal is still clearly visible. This is to say nothing of the fact that multi-dimensional colour palettes can make the graphics harder to read and hurt the accessibility of the plots (Vanderplas & Hofmann 2015).

Instead of hoping that uncertainty might collapse signal values into a single dimension, we might instead decide to do some of that work ourselves. As a matter of fact, some

uncertainty visualisation authors already have.

3.3.3 Another example: mapping combined variables

The Value Suppressing Uncertainty Palette (VSUP) (Correll et al. 2018) were designed with the intention of preventing high uncertainty values from being extracted from a map by blending colours together as they become more uncertain. Figure 4 depicts the Iowa choropleth map from Figure 3, but instead of the standard bivariate palette, we have used the VSUP approach. Since the palette was designed with the extraction of individual values in mind and it has only been tested on simple value extraction tasks (Correll et al. 2018) or search tasks (Ndlovu et al. 2023), we can see that, at least for our example, the high uncertainty spatial trend has functionally disappeared.

Finally, we have signal suppression! Well, not really, sorry, we tricked you. The two plots depicted in Figure 4 actually show the exact same data; they are both the low variance case. If you look closely, you can see that the two plots have a different scale, where plot (a) has been scaled according to our existing knowledge about this data, while plot (b) has been scaled using the range of the data passed to the plot. Since the variance and estimate are scaled independently, arbitrary differences in the range of our variance, unrelated to the estimate itself, will have significant impacts on the visual appearance of our plot. The scale issue in VSUP maps was also recognised by Kay (2019), who noted that the suppression of any one hypothesis largely depends on the methods we use to combine the palette, and the variance levels at which the blending occurs. This means that, for us to know that our plot will successfully perform signal suppression, we need to already know what signal we are trying to suppress and set up the VSUP palette accordingly. This means that VSUP maps are not suitable for exploratory data analysis.

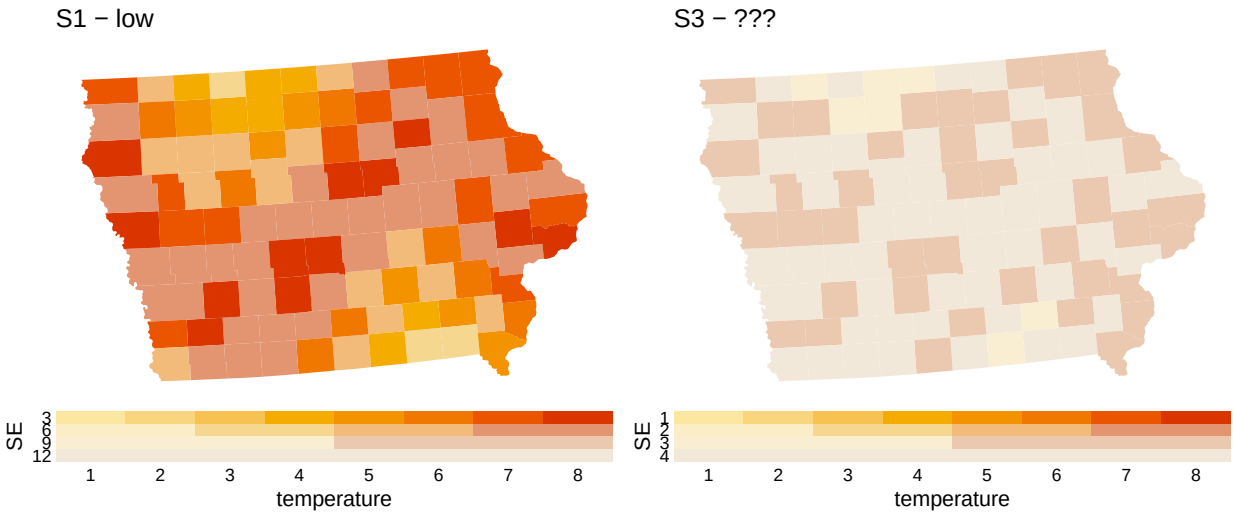

Figure 4: A variation of the choropleth map designed with a VSUP colour palette. The counties of Iowa are coloured acording to its average daily temperature and the variance in temperature. Similar to the bivariate map, temperature is mapped to hue while variance is mapped to saturation. Unlike the bivariance map, the colour space we are mapping our variables to has been transformed so that high variance estimates are harder to discern from each other. One of the maps has successfully implemented signal suppression, but a closer look at the scale brings into question if this should be the case.

3.3.4 Uncertainty and exploratory data analysis

The reason there is no uncertainty in descriptive statistics is not due to a unique property of descriptive statistics, but rather due to the lack of inference. Descriptive statistics are actually a small piece of a much larger subfield that does not perform statistical inference, and that subfield is exploratory data analysis (EDA). Tukey et al. (1977) described EDA as the process of searching for interesting hypotheses, and defined it in relation to confirmatory data analysis (CDA), the process of verifying a hypothesis. This EDA vs CDA dichotomy is the same dichotomy we drew earlier between descriptive and inferential statistics. Some have argued that the two methods are not entirely distinct, as it is often difficult to draw a hard line between EDA and CDA (Hullman & Gelman 2021). As with many concepts, these approaches exist on a continuum, where we have an inherent trade-off between the number of hypotheses we can look for and the certainty of any conclusions reached. It can help to think of the knowledge-generating process of EDA and CDA as the nozzle on a hose with multiple spray options, where EDA is a fine misting spray that touches everything in the room, and CDA is a high-pressure jet capable of obliterating any and all debris from any single spot.

The EDA versus CDA dichotomy can create some confusion when it comes to understanding the source of uncertainty in our analysis. This is why, despite the fact that an uncertainty visualisation system for EDA is one of the most discussed topics in the field (Hadjimichael et al. 2024, Peña-Araya et al. 2025, MacEachren et al. 2005, Sarma et al. 2024, Griethe & Schumann 2006), we have avoided the topic until now. The EDA versus CDA dichotomy can be compared to the dichotomy between induction, for building theories, and deduction, for testing theories, from formal logic. One of the trade-offs in the two methods is that deductive conclusions provide certainty, while conclusions from induction are inherently uncertain.

This means that EDA, the inductive counterpart, has uncertainty in any conclusions reached through the method, with the requirement to follow up our newfound hypothesis with CDA if we want true certainty. This adds another layer of confusion to our analysis, as authors frequently conflate the uncertainty in our data with the uncertainty that is inherent to EDA itself.

Authors often try to account for this inherent uncertainty and believe it should be incorporated into their plots, an argument made explicitly by Hullman & Gelman (2021) when they said there is no such thing as a "model-free" visualisation. The fear of this inherent uncertainty leads authors to try to perform some kind of statistical super-test, where they pre-emptively hedge against every false inference that could possibly be drawn from a graphic (Guo et al. 2024, Savvides et al. 2019, Hullman & Gelman 2021). This is typically done by visualising a null distribution alongside the observed data on a single plot, an approach that is statistically identical to the line-up protocol (Buja et al. 2009). As a line-up protocols were designed to a graphical parallel to standard statistical hypothesis tests, a core tool of inductive statistics, they are deductive reasoning, and therefore not EDA. The sum total of a lot of CDA is not EDA, just as frantically swinging a high-pressure jet around a room is not equivalent to using a misting spray. Even if you do manage to wet everything, you are going to use an exorbitant amount of water to do it (and possibly also break a few things). This is not to say there is anything wrong with these approaches; they are incredibly useful tools for induction. The line-up protocol, in particular, is a useful alternative to hypothesis testing, as it allows for statistical inference when there is no classical test available or when tests involve multiple effects (Hofmann et al. 2012). Rather, the problem is that we cannot simultaneously perform EDA and CDA as the approaches are, philosophically speaking, perpendicular to one another.

To truly create an uncertainty visualisation approach that is capable of EDA, we need to accept that uncertainty is inherent to the method, and it cannot be pre-emptively removed from the visualisations. If we accept this fact, then the only possible source of uncertainty in an uncertainty visualisation system that performs EDA is from the data itself. That is to say, the only possible way for there to be uncertainty in a visualisation designed for EDA is that the data itself represents inference that has been done earlier in our analysis. We can see this in the original description of Figure 1 where the uncertainty in all cases, even measurement error, represents inference that was performed earlier in our analysis.

In the VSUP approach, our ability to arbitrarily decide which values to blend at, or which suppression approach to use, means that the "uncertainty" we are visualising will be informed by the conclusions we are drawing, and not a product of the data itself. This makes the approach antithetical to EDA. For a visualisation to be suitable for EDA, it should always look the same regardless of what hypothesis we plan to draw from it. As much of the transparency in data visualisation comes from this feature in EDA, it is reasonable to set it as a requirement of our visualisations. Ensuring this property means we cannot treat signal and noise as separate variables, but rather as a single integrated unit, the yin and yang of our analysis.

3.4 Uncertainty as a distribution

Rather than trying to summarise our random variables down into a single value or pair of values, why don't we just visualise the whole distribution? This is also a common sentiment in the literature, and we can see the approach in computer science with the HOPs (Hullman et al. 2015), in geoscience with pixel-maps (Lucchesi et al. 2021, Blenkinsop et al. 2000), and in statistics with spaghetti plots.

3.4.1 Example: visualising samples

Figure 3 depicts the six visualisations Figure 1 as distributions represented by either a sample of outcomes or a quantile dot plot. Plots (a) and (b) show a linear regression as a sample of possible outcomes from the distribution. Plots (c) and (d) show a pixel map, which is a choropleth map with each county coloured by a sample of outcomes from the distribution. Plots (e) and (f) show each univariate distribution as a quantile dot plot. We can see that the strong downward trend in the linear regression, the sine wave in the choropleth map, and the incremental increase in the univariate distributions are all clearly visible in the low variance case, and invisible in the high variance cases. This means the graphic has finally achieved all criteria for signal suppression. In addition to successfully performing signal suppression, visualising the random variable as a distribution gives us additional information, such as the previously hidden bimodality of the univariate distributions in (e) and (f).

3.4.2 Quantified versus unquantified uncertainty

The visualisation in Figure 5 performs signal suppression, without sacrificing the viewer's ability to extract general statistics, unless those statistics *should* be harder to extract due to the uncertainty in the value. By showing our distribution as "data", we are able to read the "uncertainty" plots using the same perceptual tasks that are used to read the "non-uncertainty" plot. This leads to more effective communication, as people tend to read more complicated visualisations like the bivariate and VSUP plots the same way they read the simple choropleth counterpart (Ndlovu et al. 2023). This approach also does not significantly hinder our ability to extract the individual values mapped by the previous plots, as extracting global statistics from a sample can be done with relative ease (Franconeri

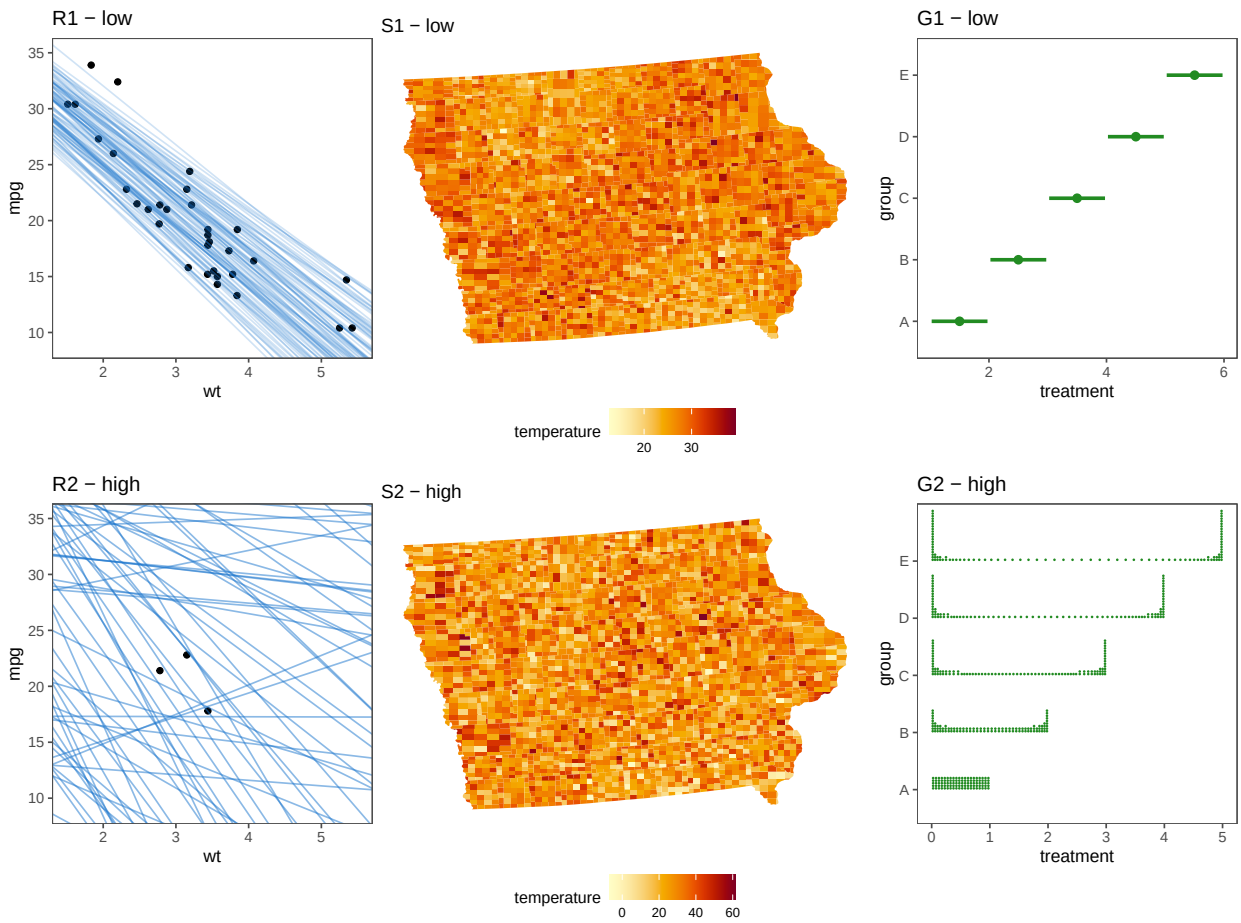

Figure 5: Six different plots, each depicting a high and low uncertainty case for our three examples: regression, choropleth, and group comparison. Plots (a) and (b) show a linear regression as a sample of possible outcomes from the distribution. Plots (c) and (d) show a pixel map, which is a choropleth map with each county coloured by a sample of outcomes from the distribution. Plots (e) and (f) show the groups as quantile dot plots. We can see that the strong downward trend in the linear regression, the sine wave in the choropleth map, and the incremental increase in the groups have all disappeared.

2021). We are able to include more information by offloading more computation to visual processing, but what is the limitation on this approach? Which aspects of uncertainty should be processed by our visual system, and which should be processed by our statistical computation? It is not obvious from the question, but this is actually a question about how much of our uncertainty should be quantified.

Quantified uncertainty usually focuses narrowly on concepts such as probability, confidence intervals, variance, error, or precision (Hullman et al. 2018, Maceachren et al. 2012, Thomson et al. 2005), while unquantified uncertainty often includes a broader range of concepts like missing values, reliability, model validity, or source integrity (Griethe & Schumann 2006, Wilkinson 2005, Pang et al. 1997, Pham et al. 2009, Boukhelifa et al. 2017). When discussing uncertainty, we typically include these unquantified uncertainties, not because these things *are* uncertainty, but because they can *create* uncertainty when we perform inference. This is often because these unquantified uncertainties violate our assumptions of the uniformity of nature (Otsuka 2023).

Sometimes we are able to visualise these assumption violations directly. For example, we can check for structure in our missing data using the `naniar` package Tierney & Cook (2023) that allows us to include missing values as a "shadow" alongside our usual visualisations. This approach amounts to just "showing the data", which is a simple but effective option for uncertainty visualisation that is largely overlooked. While this approach is useful for better understanding data, it will not eliminate trends that have become invalid due to structure in missing data or an invalid model. We can only integrate uncertainty as noise when that uncertainty has been *quantified* as an effect on the estimates we are visualising. This is not to say you should do one method over another; visualising both quantified and unquantified uncertainty is necessary for a healthy analysis. Data analysis often works in cycles, where

we find assumption violations using EDA, quantify the effect of these violations on our inference, and then visualise the output of that inference using uncertainty visualisation.

4 Evaluating uncertainty visualisations

If we want to make conclusions about how effective any uncertainty visualisation method is, we need to look at the results of evaluation experiments. The illustrative examples we have used thus far, i.e. showing a graphic and saying "wow, look at this", are lacking if we want results that should be taken seriously. Unfortunately, the conflicting results in the field are not limited to plot design and extend to the experimental findings as well (MacEachren et al. 2005, Kinkeldey et al. 2014, Hullman 2016). There are as many explanations for the noisy evaluation studies as there are contradictions in the research itself. Kim et al. (2019) believes there is some interference in results from participants' prior beliefs; Hullman (2016) believes the noise in the literature could come from visual heuristics, subjective probabilities, unknown participant utility functions, or a misunderstanding of statistical concepts (such as confidence intervals). Kinkeldey et al. (2014) suggest the best visualisation changes depending on the audience, so we cannot expect the same results between different subpopulations. Brennen & Tuerk (2018) attributes issues to cognitive load caused by the complicated nature of uncertainty visualisations, as well as the participants' prior experience in the topic. While these issues will certainly have some impact on our ability to synthesise, none of them is unique to uncertainty visualisation. Rather, the issue is likely due to a disconnect between the evaluation methods used and the stated goals of any one experiment.

4.1 Current evaluation methods

The inconsistency in the broader motivations of uncertainty visualisation bleeds into the evaluation studies and creates conflicting findings. This problem is pervasive within standard visualisation evaluation literature, where studies are regularly criticised for failing to correctly align questions and plot types (Vanderplas et al. 2020), but it is particularly egregious in uncertainty visualisation because, unlike other statistics, uncertainty plays two roles in our analysis. The role of signal is to be visible, while the role of noise is to obfuscate, so flipping the role of uncertainty also flips our conclusions about a method's effectiveness. This problem was also identified by Brennen & Tuerk (2018), their solution was to measure cognitive load and confidence, which would not alleviate the root cause of this issue. This complication is further aggravated by the fact that authors often provide little justification for their chosen response models (Hullman 2016), and the fact that approaches are compared on the basis of being "uncertainty visualisations" (Ibrekk & Morgan 1987, Hullman et al. 2015, Hofman et al. 2020), a classification that holds little to no meaning.

4.1.1 Value Extraction

Uncertainty visualisations are most commonly evaluated based on how accurately viewers can extract an estimate and its variance (Kinkeldey et al. 2014, Hullman et al. 2019). This is not unusual, as direct observation is the simplest way to verify that information can be accurately read from a graph (Vanderplas et al. 2020). Unfortunately, this approach doesn't work for uncertainty visualisation. The second we ask a specific question about a statistic, that statistic becomes our inferential statistic, even if the plot was not the intent behind the question. By shifting the focus from $\hat{X}$ to $Var(\hat{X})$ or $P(\hat{X} < x)$, we end up evaluating visualisations on their ability to convey uncertainty statistics, rather

than their ability to perform signal suppression. There is nothing unique to uncertainty in these studies, so when we boil the findings down to generalised results, they simply restate existing principles within information visualisation, even if they do not realise it themselves. Some of the findings are particularly obvious, such as studies that found participants were more accurate when reading a probability expressed as text than when they had to extract it from a graphic (Cheong et al. 2016, Savelli & Joslyn 2013). Other studies replicate existing research, such as the finding that a probability mapped to a position is more accurate than one mapped to an area (Ibrekk & Morgan 1987, Gschwandtner et al. 2016), a finding that established the hierarchy of perceptual tasks (Cleveland & McGill 1984, Vanderplas et al. 2020), research dating back over 40 years. This extends beyond simple accuracy evaluations, as some authors have found that colour was more effective than size when searching for maximums and minimums in variances (Sanyal et al. 2009), even though it has been well known that pre-attentive aesthetics, such as colour, are more efficient for search tasks since the 1980s (Vanderplas et al. 2020). By classifying these studies as evaluations of "uncertainty visualisation" while evaluating uncertainty as a signal, we are encouraged to see successful examples of signal suppression as failure. This approach leads authors to advise against particular aesthetic mappings for uncertainty, because they cause participants to have more difficulty extracting values (Blenkinsop et al. 2000). This conclusion is antithetical to the goals of signal suppression and occurs because these methods evaluate uncertainty as a signal, not as noise.

4.1.2 Trust, confidence, and risk aversion

If we cannot directly measure uncertainty for fear that it turns into a signal, we might then assume we can measure the secondary benefits of increased transparency. This seems to be the approach of many visualisation authors, as secondary benefits such as trust, confidence,

and risk aversion are all frequently used in uncertainty evaluation studies (Hullman et al. 2019). Unfortunately, measuring these secondary effects often leads to confusing conclusions that simultaneously argue for and against the inclusion of uncertainty.

This is most commonly noticed in the use of trust as a measure, as several authors have commented that measuring trust, and not transparency, can lead to a questionable subtext that argues against transparency (Spiegelhalter 2017, O'Neill 2018). We see this directly play out in the visualisation literature, where surveyed visualisation authors explicitly said they didn't include uncertainty due to the fact that they might decrease trust in their conclusions (Hullman 2020). This sentiment is also true for confidence, as Blenkinsop et al. (2000) commented that visually integrable depictions of uncertainty should be avoided, as they decrease the viewer's confidence in their extracted data values. Confidence is also used as a proxy for poorly constructed visualisation in standard visualisation experiments, so using confidence as a measure of uncertainty collapses these confounding factors down into a single metric.

Another secondary effect that is similar to trust and confidence is risk aversion. Risk aversion is an economic term used to describe an agent who would choose a random variable with a lower expected payout because it also has a lower variance. Risk aversion's role in the uncertainty visualisation is unclear, as authors will argue uncertainty should elicit more risk aversion in one paper (Hullman et al. 2019), and argue for less risk aversion (by proxy of suggesting rational agents as a benchmark) in the next (Wu et al. 2023). Ultimately, these approaches have similar issues to the value extraction studies, except they are slightly more confusing in their goals, leading them to simultaneously argue for and against the inclusion of uncertainty in a visualisation.

4.1.3 Vague Questions

Some authors recognise these issues in the evaluation literature, but their attempts to evaluate uncertainty as noise are often ad hoc with varying levels of success. Some authors try to incorporate uncertainty by asking vague questions but compare the results to a ground truth, creating a situation where plots are *evaluated* as a value extraction study without participants being asked the *questions* of a value extraction task. Examples include asking participants for the "best estimate" (Ibrekk & Morgan 1987), or identifying when a distribution is "furthest to the right" (Hofmann et al. 2012) and then comparing these results to the distribution's expected value. The convoluted questions also seem to frequently confuse the participants and lead to obviously wrong responses. For example, Hullman et al. (2015) asked participants to compare two normally distributed groups, A and B, and had many participants say that group A was more likely to be bigger, despite group B having a higher mean. In another study, Gschwandtner et al. (2016) asked participants the "probability that the interval has already ended at the marked point in time?" and participants replied with the probability that the interval had already started. The confusion around trying to capture the effects of uncertainty can also extend to the authors of the study itself. We can see an example of this in Padilla et al. (2017) where in order to answer the question correctly, the first experiment required participants to assume that an oil rig being"more likely to be hit" by a hurricane would *not* translate to the rig sustaining "more damage", while the second experiment required participants to assume the opposite. This contradictory assumption in how to appropriately use uncertainty information seemed to go unnoticed by the authors. While these approaches are certainly a step in the right direction, the experiments overcomplicate the problem and ultimately confuse everyone involved.

4.1.4 Effective methods

This is not to say all evaluation studies fail to properly evaluate uncertainty as noise. There are several studies that ask participants to identify a particular signal that the noise is trying to obfuscate (Kale et al. 2018, Correll & Gleicher 2014), which seems to be an effective method. The only problem with these studies is that most uncertainty visualisation methods are not integrated into *the grammar of graphics* (Wilkinson 2005, Wickham 2010) so we regularly see comparisons between disparate plots that would never be considered substitutes for one another outside the artificial "uncertainty visualisation" framework they are placed within. For example, Kale et al. (2018) compared static bar charts with error bars to a bar chart with animated samples, meaning that any difference in participants' ability to read the plot could be due to the statistic (confidence interval versus sample), the geometry (bars versus intervals), or the use of animation (static versus animated plots). This means that even when evaluations *are* done correctly, there is no generalisable theory we can take from the results.

The point here is not to mock the existing literature for poor academic rigour. If you were to go read these papers yourself, you would find them to (usually) be logically consistent and well-formulated pieces of work. Rather, the point is to illustrate that evaluating uncertainty as noise is surprisingly difficult. Designing tests for signal suppression will require a formalisation of uncertainty within the grammar of graphics, as well as improved evaluation methods.

4.2 Implicit Hypothesis Testing

The main problems with current uncertainty visualisation evaluations are that they often require explicit (or convoluted) questions about the variance. Asking direct questions about

the statistics or outcomes is not an explicit requirement of visualisation evaluations. In their review of testing statistical graphics, Vanderplas et al. (2020) drew a distinction between explicit tests, where participants are asked direct questions about specific features of a plot, and implicit testing, where users identify both the purpose and function of the plot. The line-up protocol is the most salient example of the implicit approach. Line-ups are a confirmatory visualisation tool where participants are shown a set of $M - 1$ plots, and asked to identify the plot that is the "most different", leaving participants to decide what "most different" means to them, even if it is not what the authors intended (VanderPlas & Hofmann n.d.). The implicit testing does not limit the versatility of the approach, with the line-up being used to evaluate the effectiveness of different types of plots (Hofmann et al. 2012), colour palettes (Reda & Szafir 2021), and design decisions (VanderPlas & Hofmann n.d.).

Line-up protocols are not only useful for their implicit testing, but they also have parallels to hypothesis testing that can be leveraged for uncertainty visualisation. The concept of signal suppression is, at its core, an assertion of statistical validity, which means the visibility of signals should be directly proportional to p-values or some equivalent measure. This comparison is not new in uncertainty visualisation, where parallels have been drawn to frequentist statistics by Correll & Gleicher (2014), who compared results to Cohen's D, and to Bayesian statistics by Kim et al. (2019), who evaluated plots based on their impact on users' prior beliefs. The comparison to hypothesis testing is far more natural for the line-up protocol, with Majumder et al. (2013) explicitly defining a visual test statistic, while Li et al. (2024) compared line-ups to standard statistical tests using power curves. The parallels to line-ups and uncertainty visualisation are numerous, with the relationship even being identified by Hullman et al. (2015) in their development of hypothetical outcome plots.

The line-up protocol and uncertainty visualisation share a lot in common, with the motivation behind line-ups also rooted in checking if perceived patterns are real or merely the result of chance (Buja et al. 2009, Wickham et al. 2010). As both approaches are attempting to do the same thing, it is likely that we are unable to leverage the line-up protocol directly to evaluate uncertainty visualisation, but a new evaluation methodology should try to learn from the success of the line-up approach. Designing an implicit testing method for uncertainty visualisation that allows us to draw parallels to standard notions of statistical significance would solve many of the issues with the current evaluation approaches.

4.3 Heuristics research

There is one element of uncertainty visualisation research that we have not touched on yet, but it is incredibly consistent, and that is the research around visual heuristics. The heuristics research, unlike the plot evaluation work, is incredibly consistent, likely because heuristics are indifferent to the goal of our analysis and uncertainty's role as signal or noise. Several heuristics are of particular importance for uncertainty visualisation. We will give a brief overview of them here, and for a more in-depth analysis, see Padilla, Kay & Hullman (2022).

The use of confidence intervals or boundaries has a few known issues in their interpretation. The sine illusion can cause the confidence interval of a smoothed sine curve to seem wider at the peaks than the troughs, causing us to underestimate uncertainty associated with changing values (Vanderplas & Hofmann 2015). Similarly, researchers frequently find that visual boundaries such as the cone of uncertainty in hurricane forecasts (Padilla, Kay & Hullman 2022), or confidence bands or ellipses (VanderPlas & Hofmann n.d.) create the impression of a meaningful bound, separating points outside the bounds from those within

it, which can create problems if said bound is not inherently meaningful.

While many researchers tend to prefer frequency framing, they are not without their issues. Points that were on an outcome of an ensemble display were perceived as more likely than points not on an outcome, even when the point that was not on an outcome was closer to the centre of the distribution (and therefore more likely) (Padilla et al. 2017). This is considered to be a variation of the within-bar bias, where outcomes within the bar on a bar chart as perceived to be more likely than those outside it (Newman & Scholl 2012).

We should also try to avoid visualising central estimates if possible, as several studies have found that viewers ignore uncertainty information and compare the difference between the two means if they have the option (Padilla, Kay & Hullman 2022).

These heuristics have the potential to create noise in signal suppression evaluations, and should be kept in mind when designing uncertainty visualisations as well as evaluation experiments.

5 Conclusions and Future Work

This paper has formalised the purpose of uncertainty visualisation, and in doing so, highlighted several gaps in the existing literature.

Experimental practices on uncertainty visualisation need to be standardised. Some existing evaluation experiments treat uncertainty as a signal, while others treat uncertainty as noise. As the literature currently exists, there is no way to combine papers to get a meaningful sense of how uncertainty information is understood by a viewer. Researchers need to ensure that when they identify the motivation behind their visualisation technique, their evaluation methods align with the stated goals of the paper.

Additionally *experimental methods that evaluate uncertainty as noise need to be developed.* Research into separability and integrability is of particular interest to uncertainty visualisation, as it allows interference from the uncertainty variable. When designing experiments, authors often choose aesthetics that are visually distinguishable; uncertainty visualisation authors should consider doing the opposite.

Uncertainty needs to be formalised within the grammar of graphics. The ad hoc nature of the field is in desperate need of formalisation. Some of this formalisation was done by Kay (2023), but this work was limited to the visualisation of univariate distributions. Giving authors the ability to describe uncertainty visualisations in terms of statistics, geometries and aesthetics will support evaluation experiments that can build towards a cohesive theory of visualising uncertainty as noise.

Software that allows users to easily perform signal suppression should be developed. Existing uncertainty visualisation methods view a distribution as its own object, and there are no software options for the "an uncertainty visualisation is a function of an existing visualisation" philosophy.

Signal suppression is an untouched area of visualisation research, and developing methods for the practice may require us to challenge our entire notion of what makes a good visualisation. We are interested in seeing how this challenge plays out.

6 Acknowledgements

The first author of this paper is supported in part by a scholarship from the Australian Energy Market Operator. This research was supported by the Commonwealth through an Australian Government Research Training Program Scholarship [DOI: https://doi.org/10.82133/C42F-

K220]. The R packages were used for this work were: `tidyverse` (Wickham et al. 2019), `RColorBrewer` (Neuwirth 2022), `scales` (Wickham et al. 2023), `sf` (Pebesma & Bivand 2023), `urbnmapr` (Strochak et al. 2024), `flextable` (Gohel & Skintzos 2024), `colorspace` (Stauffer et al. 2009), `ggdist` (Kay 2023), `ggdibbler` (Mason et al. 2026), `patchwork` (Pedersen 2025), `distributional` (O'Hara-Wild et al. 2024), `ggthemes` (Arnold 2024), and `rgeos` (Bivand & Rundel 2023). The GitHub repository for this paper can be found at https://github.com/harriet-mason/ARSA-UncertaintyLitReview, which contains the files required to reproduce this article in full.

Bibliography

Anscombe, F. J. (1973), 'Graphs in statistical analysis', *The American Statistician* **27**(1), 17–21.

URL: *https://www.tandfonline.com/doi/abs/10.1080/00031305.1973.10478966*

Arnold, J. B. (2024), *ggthemes: Extra Themes, Scales and Geoms for 'ggplot2'*. R package version 5.1.0.

URL: *https://CRAN.R-project.org/package=ggthemes*

Begg, S. H., Welsh, M. B. & Bratvold, R. B. (2014), Uncertainty vs. variability: What's the difference and why is it important?, *in* 'SPE Hydrocarbon Economics and Evaluation Symposium', Vol. SPE Hydrocarbon Economics and Evaluation Symposium.

URL: *https://doi.org/10.2118/169850-MS*

Benjamin, D. M. & Budescu, D. V. (2018), 'The role of type and source of uncertainty on the processing of climate models projections', *Frontiers in Psychology* **9**(MAR), 1–17.

Bivand, R. & Rundel, C. (2023), *rgeos: Interface to Geometry Engine - Open Source*

('GEOS'). R package version 0.6-3.

URL: *https://CRAN.R-project.org/package=rgeos*

Blenkinsop, S., Fisher, P., Bastin, L. & Wood, J. (2000), 'Evaluating the perception of uncertainty in alternative visualization strategies', *Cartographica* **37**(1), 1–13.

Boukhelifa, N., Bezerianos, A., Isenberg, T. & Fekete, J. D. (2012), 'Evaluating sketchiness as a visual variable for the depiction of qualitative uncertainty', *IEEE Transactions on Visualization and Computer Graphics* **18**(12), 2769–2778.

Boukhelifa, N., Perrin, M. E., Huron, S. & Eagan, J. (2017), 'How data workers cope with uncertainty: A task characterisation study', *Conference on Human Factors in Computing Systems - Proceedings* **2017-May**(May), 3645–3656.

Brennen, A. & Tuerk, S. (2018), 'An instrument for evaluating uncertainty visualization techniques', *Conference on Human Factors in Computing Systems - Proceedings* **2018-April**, 1–6. ISBN: 9781450356206.

Buja, A., Cook, D., Hofmann, H., Lawrence, M., Lee, E. K., Swayne, D. F. & Wickham, H. (2009), 'Statistical inference for exploratory data analysis and model diagnostics', *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences* **367**(1906), 4361–4383.

Chakraborty, S., Kiefer, P. & Raubal, M. (2024), 'The influence of uncertainty visualization on cognitive load in a safety- and time-critical decision-making task', *International Journal of Geographical Information Science* **38**(8), 1583–1610.

URL: *https://www.tandfonline.com/doi/full/10.1080/13658816.2024.2348747*

Cheong, L., Bleisch, S., Kealy, A., Tolhurst, K., Wilkening, T. & Duckham, M. (2016), 'Evaluating the impact of visualization of wildfire hazard upon decision-making under

uncertainty', *International Journal of Geographical Information Science* **30**(7), 1377–1404.
URL: *http://dx.doi.org/10.1080/13658816.2015.1131829*

Cleveland, W. S. & McGill, R. (1984), 'Graphical perception: Theory, experimentation, and application to the development of graphical methods', *Journal of the American Statistical Association* **79**(387), 531–554.

Correll, M. & Gleicher, M. (2014), 'Error bars considered harmful: Exploring alternate encodings for mean and error', *IEEE Transactions on Visualization and Computer Graphics* **20**(12), 2142–2151.

Correll, M., Moritz, D. & Heer, J. (2018), 'Value-suppressing uncertainty palettes', *Conference on Human Factors in Computing Systems - Proceedings* **2018-April**, 1–11.

Fischhoff, B. & Davis, A. L. (2014), 'Communicating scientific uncertainty', *Proceedings of the National Academy of Sciences of the United States of America* **111**, 13664–13671.

Franconeri, S. L. (2021), 'Three perceptual tools for seeing and understanding visualized data', *Current Directions in Psychological Science* **30**(5), 367–375.

Gohel, D. & Skintzos, P. (2024), *flextable: Functions for Tabular Reporting.* R package version 0.9.6.
URL: *https://CRAN.R-project.org/package=flextable*

Griethe, H. & Schumann, H. (2006), The visualization of uncertain data: Methods and problems, *in* 'SimVis', Vol. 6, pp. 143–156.

Gschwandtner, T., Bögl, M., Federico, P. & Miksch, S. (2016), 'Visual encodings of temporal uncertainty: A comparative user study', *IEEE Transactions on Visualization and Computer Graphics* **22**(1), 539–548.

Guo, Z., Kale, A., Kay, M. & Hullman, J. (2024), 'Vmc: A grammar for visualizing statistical model checks', *IEEE Transactions on Visualization and Computer Graphics* .

Gustafson, A. & Rice, R. E. (2019), 'The effects of uncertainty frames in three science communication topics', *Science Communication* **41**(6), 679–706.

Hadjimichael, A., Schlumberger, J. & Haasnoot, M. (2024), 'Data visualisation for decision making under deep uncertainty: current challenges and opportunities', *Environmental Research Letters* **19**(11), 111011.

URL: *https://iopscience.iop.org/article/10.1088/1748-9326/ad858b*

Hofman, J. M., Goldstein, D. G. & Hullman, J. (2020), 'How visualizing inferential uncertainty can mislead readers about treatment effects in scientific results', *Conference on Human Factors in Computing Systems - Proceedings* .

Hofmann, H., Follett, L., Majumder, M. & Cook, D. (2012), 'Graphical tests for power comparison of competing designs', *IEEE Transactions on Visualization and Computer Graphics* **18**(12), 2441–2448.

Hullman, J. (2016), 'Why evaluating uncertainty visualization is error prone', *ACM International Conference Proceeding Series* **24-October**, 143–151.

Hullman, J. (2020), 'Why authors don't visualize uncertainty', *IEEE Transactions on Visualization and Computer Graphics* **26**(1), 130–139.

Hullman, J. & Gelman, A. (2021), 'Designing for interactive exploratory data analysis requires theories of graphical inference', *Harvard Data Science Review* pp. 1–70.

Hullman, J., Kay, M., Kim, Y. S. & Shrestha, S. (2018), 'Imagining replications: Graphical

prediction discrete visualizations improve recall estimation of effect uncertainty', *IEEE Transactions on Visualization and Computer Graphics* **24**(1), 446–456.

Hullman, J., Qiao, X., Correll, M., Kale, A. & Kay, M. (2019), 'In pursuit of error: A survey of uncertainty visualization evaluation', *IEEE Transactions on Visualization and Computer Graphics* **25**(1), 903–913.

Hullman, J., Resnick, P. & Adar, E. (2015), 'Hypothetical outcome plots outperform error bars and violin plots for inferences about reliability of variable ordering', *PLoS ONE* **10**(11).

Ibrekk, H. & Morgan, M. G. (1987), 'Graphical communication of uncertain quantities to nontechnical people', *Risk Analysis* **7**(4), 519–529.

Kale, A., Kay, M. & Hullman, J. (2021), 'Visual reasoning strategies for effect size judgments and decisions', *IEEE Transactions on Visualization and Computer Graphics* **27**(2), 272–282.

Kale, A., Nguyen, F., Kay, M. & Hullman, J. (2018), 'Hypothetical outcome plots help untrained observers judge trends in ambiguous data', *IEEE transactions on visualization and computer graphics* **25**(1), 892–902.

Kay, M. (2019), 'How much value should an uncertainty palette suppress if an uncertainty palette should suppress value? statistical and perceptual perspectives'.

Kay, M. (2023), 'ggdist: Visualizations of distributions and uncertainty in the grammar of graphics', *IEEE Transactions on Visualization and Computer Graphics* **30**(1), 414–424.

Kim, Y. S., Walls, L. A., Krafft, P. & Hullman, J. (2019), 'A bayesian cognition approach

to improve data visualization', *Conference on Human Factors in Computing Systems - Proceedings* pp. 1–14.

Kinkeldey, C., MacEachren, A. M. & Schiewe, J. (2014), 'How to assess visual communication of uncertainty? a systematic review of geospatial uncertainty visualisation user studies', *Cartographic Journal* **51**(4), 372–386.

Koonchanok, R., Tawde, G. Y., Narayanasamy, G. R., Walimbe, S. & Reda, K. (2023), 'Visual belief elicitation reduces the incidence of false discovery', *Conference on Human Factors in Computing Systems - Proceedings* .

Kuhnert, P. M., Pagendam, D. E., Bartley, R., Gladish, D. W., Lewis, S. E. & Bainbridge, Z. T. (2018), 'Making management decisions in the face of uncertainty: A case study using the burdekin catchment in the great barrier reef', *Marine and Freshwater Research* **69**(8), 1187–1200.

Lee, C., Yang, T., Inchoco, G. D., Jones, G. M. & Satyanarayan, A. (2021), Viral visualizations: How coronavirus skeptics use orthodox data practices to promote unorthodox science online, *in* 'Proceedings of the 2021 CHI conference on human factors in computing systems', pp. 1–18.

Li, W., Cook, D., Tanaka, E. & VanderPlas, S. (2024), 'A plot is worth a thousand tests: Assessing residual diagnostics with the lineup protocol', *Journal of Computational and Graphical Statistics* pp. 1–19.

Locke, S. & D'Agostino McGowan, L. (2018), *datasauRus: Datasets from the Datasaurus Dozen.* R package version 0.1.4.

URL: *https://CRAN.R-project.org/package=datasauRus*

Lucchesi, L., Kuhnert, P. & Wikle, C. (2021), 'Vizumap: an R package for visualising uncertainty in spatial data', *Journal of Open Source Software* **6**(59), 2409.

MacEachren, A. M. (1992), 'cartographic perspectives visualizing uncertain information', *Cartographic perspectives* **13**, 10–19.

MacEachren, A. M., Robinson, A., Hopper, S., Gardner, S., Murray, R., Gahegan, M. & Hetzler, E. (2005), 'Visualizing geospatial information uncertainty: What we know and what we need to know', *Cartography and Geographic Information Science* **32**(3), 139–160. ISBN: 1523040054738.

Maceachren, A. M., Roth, R. E., O'Brien, J., Li, B., Swingley, D. & Gahegan, M. (2012), 'Visual semiotics & uncertainty visualization: An empirical study', *IEEE Transactions on Visualization and Computer Graphics* **18**(12), 2496–2505.

Majumder, M., Hofmann, H. & Cook, D. (2013), 'Validation of visual statistical inference, applied to linear models', *Journal of the American Statistical Association* **108**(503), 942–956.
URL: *http://www.tandfonline.com/doi/abs/10.1080/01621459.2013.808157*

Manski, C. F. (2020), 'The lure of incredible certitude', *Economics and Philosophy* **36**(2), 216–245.

Mason, H., Cook, D., Goodwin, S. & VanderPlas, S. (2026), *ggdibbler: Add Uncertainty to Data Visualisations.* R package version 0.6.4.9000, commit 2d3c25a6147e1bcc36d26d8c1b99a3442198d2ca.
URL: *https://github.com/harriet-mason/ggdibbler*

Matejka, J. & Fitzmaurice, G. (2017), Same stats, different graphs: Generating datasets with varied appearance and identical statistics through simulated annealing, *in* 'Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems', CHI '17, Association

for Computing Machinery, New York, NY, USA, p. 1290–1294.

URL: *https://doi.org/10.1145/3025453.3025912*

Meng, X. L. (2014), 'A trio of inference problems that could win you a nobel prize in statistics (if you help fund it)', *Past, Present, and Future of Statistical Science* pp. 537–562.

Moritz, D., Fisher, D., Ding, B. & Wang, C. (2017), Trust, but verify: Optimistic visualizations of approximate queries for exploring big data, *in* 'Proceedings of the 2017 CHI conference on human factors in computing systems', pp. 2904–2915.

Ndlovu, A., Shrestha, H. & Harrison, L. T. (2023), Taken by surprise? evaluating how bayesian surprise & suppression influences peoples' takeaways in map visualizations, *in* '2023 IEEE Visualization and Visual Analytics (VIS)', IEEE, pp. 136–140.

Neuwirth, E. (2022), *RColorBrewer: ColorBrewer Palettes*. R package version 1.1-3.

URL: *https://CRAN.R-project.org/package=RColorBrewer*

Newman, G. E. & Scholl, B. J. (2012), 'Bar graphs depicting averages are perceptually misinterpreted: The within-the-bar bias', *Psychonomic Bulletin and Review* **19**(4), 601–607.

O'Hara-Wild, M., Kay, M., Hayes, A. & Hyndman, R. (2024), *distributional: Vectorised Probability Distributions*. R package version 0.5.0.

URL: *https://CRAN.R-project.org/package=distributional*

Olston, C. & Mackinlay, J. D. (2002), 'Visualizing data with bounded uncertainty', *Proceedings - IEEE Symposium on Information Visualization, INFO VIS* **2002-Janua**, 37–40.

O'Neill, O. (2018), 'Linking trust to trustworthiness', *International Journal of Philosophical Studies* **26**(2), 293–300.

URL: *https://doi.org/10.1080/09672559.2018.1454637*

Otsuka, J. (2023), *Thinking About Statistics: The Philosophical Foundations*, 1st ed edn, Routledge, New York.

Padilla, L., Hosseinpour, H., Fygenson, R., Howell, J., Chunara, R. & Bertini, E. (2022), 'Impact of COVID-19 forecast visualizations on pandemic risk perceptions', *Scientific Reports 2022 12:1* **12**(1), 1–14.

URL: *https://www.nature.com/articles/s41598-022-05353-1*

Padilla, L., Kay, M. & Hullman, J. (2022), *Computational Statistics in Data Science*, John Wiley & Sons, chapter 22, pp. 405–426.

Padilla, L., Powell, M., Kay, M. & Hullman, J. (2021), 'Uncertain about uncertainty: How qualitative expressions of forecaster confidence impact decision-making with uncertainty visualizations', *Frontiers in Psychology* **11**.

Padilla, L., Ruginski, I. & Creem-Regehr, S. (2017), 'Effects of ensemble and summary displays on interpretations of geospatial uncertainty data', *Cognitive Research: Principles and Implications* **2**(1).

Pang, A. T., Wittenbrink, C. M. & Lodha, S. K. (1997), 'Approaches to uncertainty visualization', *Visual Computer* **13**(8), 370–390.

Pebesma, E. & Bivand, R. (2023), *Spatial Data Science: With applications in R*, Chapman and Hall/CRC.

URL: *https://r-spatial.org/book/*

Pedersen, T. L. (2025), *patchwork: The Composer of Plots.* R package version 1.3.2.

URL: *https://CRAN.R-project.org/package=patchwork*

Peña-Araya, V., Fontaine, C. M., Wei, X., Delpech, G. & Bezerianos, A. (2025), Uncertainty

in science is malleable. advocating for user-agency in defining uncertainty in visualizations: a case study in geology, *in* 'Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems', pp. 1–18.

Pham, B., Streit, A. & Brown, R. (2009), Visualization of information uncertainty: Progress and challenges, *in* 'Advanced Information and Knowledge Processing', Vol. 36, Springer-Verlag London Ltd, pp. 19–48.

Potter, K., Kniss, J., Riesenfeld, R. & Johnson, C. R. (2010), 'Visualizing summary statistics and uncertainty', *Computer Graphics Forum* **29**(3), 823–832.

Reda, K. & Szafir, D. A. (2021), 'Rainbows revisited: Modeling effective colormap design for graphical inference', *IEEE Transactions on Visualization and Computer Graphics* **27**(2), 1032–1042.

URL: *https://ieeexplore.ieee.org/document/9222327/*

Sanyal, J., Zhang, S., Bhattacharya, G., Amburn, P. & Moorhead, R. J. (2009), 'A user study to compare four uncertainty visualization methods for 1D and 2D datasets', *IEEE Transactions on Visualization and Computer Graphics* **15**(6), 1209–1218.

Sarma, A., Guo, S., Hoffswell, J., Rossi, R., Du, F., Koh, E. & Kay, M. (2023), 'Evaluating the use of uncertainty visualisations for imputations of data missing at random in scatterplots', *IEEE Transactions on Visualization and Computer Graphics* **29**(1), 602–612.

Sarma, A., Pu, X., Cui, Y., Correll, M., Brown, E. T. & Kay, M. (2024), Odds and insights: Decision quality in exploratory data analysis under uncertainty, *in* 'Proceedings of the CHI Conference on Human Factors in Computing Systems', CHI '24, Association for Computing Machinery, New York, NY, USA.

URL: *https://doi.org/10.1145/3613904.3641995*

Savelli, S. & Joslyn, S. (2013), 'The advantages of predictive interval forecasts for non-expert users and the impact of visualizations', *Applied Cognitive Psychology* **27**(4), 527–541.

Savvides, R., Henelius, A., Oikarinen, E. & Puolamäki, K. (2019), Significance of patterns in data visualisations, *in* 'Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining', ACM, Anchorage AK USA, pp. 1509–1517.
URL: *https://dl.acm.org/doi/10.1145/3292500.3330994*

Smart, S. & Szafir, D. A. (2019), 'Measuring the separability of shape, size, and color in scatterplots', *Conference on Human Factors in Computing Systems - Proceedings* pp. 1–14.

Spiegelhalter, D. (2017), 'Risk and uncertainty communication', *Annual Review of Statistics and Its Application* **4**, 31–60.

Stauffer, R., Mayr, G. J., Dabernig, M. & Zeileis, A. (2009), 'Somewhere over the rainbow: How to make effective use of colors in meteorological visualizations', *Bulletin of the American Meteorological Society* **96**(2), 203–216.

Strochak, S., Ueyama, K. & Williams, A. (2024), *urbnmapr: State and county shapefiles in sf and tibble format*. R package version 0.0.0.9002.
URL: *https://github.com/UrbanInstitute/urbnmapr*

Thomson, J., Hetzler, E., MacEachren, A., Gahegan, M. & Pavel, M. (2005), 'A typology for visualizing uncertainty', *Visualization and Data Analysis 2005* **5669**(March 2005), 146.

Tierney, N. & Cook, D. (2023), 'Expanding tidy data principles to facilitate missing data exploration, visualization and assessment of imputations', *Journal of Statistical Software* **105**(7), 1–31.

Tukey, J. W. et al. (1977), *Exploratory data analysis*, Vol. 2, Springer.

Vanderplas, S., Cook, D. & Hofmann, H. (2020), 'Annual review of statistics and its application testing statistical charts: What makes a good graph?'.

URL: *https://doi.org/10.1146/annurev-statistics-031219-*

Vanderplas, S. & Hofmann, H. (2015), 'Signs of the sine illusion — why we need to care signs of the sine illusion — why we need to care', **8600**.

VanderPlas, S. & Hofmann, H. (n.d.), 'Clusters beat trend!? testing feature hierarchy in statistical graphics', **26**(2), 231–242.

Walker, W. E., Harremoes, P., Rotmans, J., Van Der Sluijs, J. P., Van Asselt, M. B. A., Janssen, P. & Krayer Von Krauss, M. P. (2003), 'Defining uncertainty', *Integrated Assessment* **4**(1), 5–17.

URL: *https://www.narcis.nl/publication/RecordID/oai:tudelft.nl:uuid:fdc0105c-e601-402a-8f16-ca97e9963592*

Wallsten, T. S., Budescu, D. V., Erev, I. & Diederich, A. (1997), 'Evaluating and combining subjective probability estimates', *Journal of Behavioral Decision Making* **10**(3), 243–268.

Wickham, H. (2010), 'A Layered grammar of graphics', *Journal of Computational and Graphical Statistics* **19**(1), 3–28.

URL: *https://doi.org/10.1198/jcgs.2009.07098*

Wickham, H., Averick, M., Bryan, J., Chang, W., D'Agostino McGowan, L., François, R., Grolemund, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T. L., Miller, E., Bache, S. M., Müller, K., Ooms, J., Robinson, D., Seidel, D. P., Spinu, V., Takahashi, K., Vaughan, D., Wilke, C., Woo, K. & Yutani, H. (2019), 'Welcome to the tidyverse', *Journal of Open Source Software* **4**(43), 1686.

Wickham, H., Cook, D., Hofmann, H. & Buja, A. (2010), 'Graphical inference for infovis', *IEEE Transactions on Visualization and Computer Graphics* **16**, 973–979.

Wickham, H. & Hofmann, H. (2011), 'Product plots', *IEEE Transactions on Visualization and Computer Graphics* **17**(12), 2223–2230.

Wickham, H., Pedersen, T. L. & Seidel, D. (2023), *scales: Scale Functions for Visualization*. R package version 1.3.0.
URL: *https://CRAN.R-project.org/package=scales*

Wilkinson, L. (2005), *The Grammar of Graphics (Statistics and Computing)*, Springer-Verlag, Berlin, Heidelberg.

Wu, Y., Guo, Z., Mamakos, M., Hartline, J. & Hullman, J. (2023), 'The rational agent benchmark for data visualization'.

Yang, F., Cai, M., Mortenson, C., Fakhari, H., Lokmanoglu, A. D., Hullman, J., Franconeri, S., Diakopoulos, N., Nisbet, E. C. & Kay, M. (2023), 'Swaying the public? impacts of election forecast visualizations on emotion, trust, and intention in the 2022 us midterms', *IEEE Transactions on Visualization and Computer Graphics* **30**(1), 23–33.

Zhao, J., Wang, Y., Mancenido, M. V., Chiou, E. K. & Maciejewski, R. (2023), 'Evaluating the impact of uncertainty visualization on model reliance', *IEEE Transactions on Visualization and Computer Graphics* **PP**(X), 1–15.